

COMPREHENSIVE PROJECT REPORT

Topic: Build a Security Monitoring System (AWS)

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Date: February 14, 2026

Reg No: ALT/SOE/025/4248

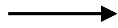
EXECUTIVE SUMMARY

This report documents the successful implementation of a comprehensive security monitoring and automated response system in AWS. The project focused on detecting unauthorized access to sensitive credentials (honeytokens) stored in AWS Secrets Manager and implementing an automated "kill-switch" to revoke access immediately upon detection.

Key Achievements:

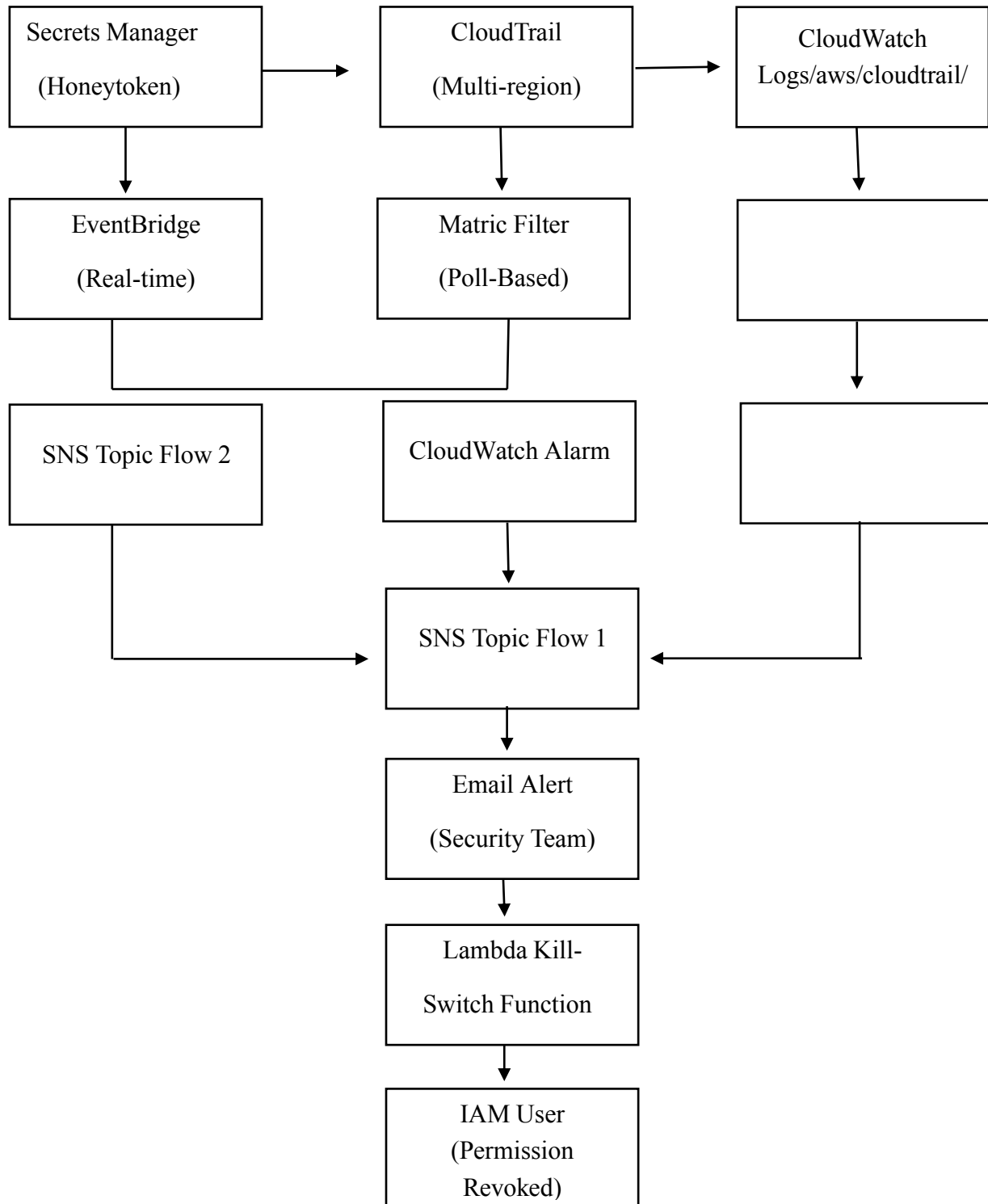
- ❖ Multi-region CloudTrail configured with CloudWatch Logs integration
- ❖ Honeytokens secret created in Secrets Manager (Production_Database_Credentials)
- ❖ Dual detection paths: CloudWatch Metric Filter (polling-based) and EventBridge (event-driven)
- ❖ Automated remediation via Lambda function that strips IAM permissions
- ❖ Successful test with "Victim" user demonstrating complete access revocation within seconds

Success Criteria Met: The system successfully detected every access attempt and automatically quarantined the offending user, preventing further API calls.



1. LAB ARCHITECTURE

1.1 High-Level Design



1.2 Core Components

Component	Purpose	Configuration Details
CloudTrail	Audit logging for all API calls	Multi-region, integrated with CloudWatch Logs
Secrets Manager	Honeytoken storage	Secret name: <i>Production_Database_Credentials</i>
CloudWatch Logs	Centralized log storage	Log group: /aws/cloudtrail/security-trail-logs
Metric Filter	Pattern matching in logs	Filter: { (\$.eventName = "GetSecretValue") && (\$.requestParameters.secretId = "Production_Database_Credentials") }
CloudWatch Alarm	Threshold-based alerting	Metric: ProductionSecretAccessCount, Threshold: ≥ 1 in 1 minute
SNS Topic (Flow 1)	Email notifications for CloudWatch	Subscription: [Student Email]
EventBridge Rule	Real-time event detection	Pattern: SecretsManager GetSecretValue events
SNS Topic (Flow 2)	Email notifications for EventBridge	Subscription: [Student Email]
Lambda Function	Automated remediation	Python 3.12, IAM permission stripping
IAM Role (Lambda)	Execution permissions	Policies: AWSLambdaBasicExecutionRole + custom inline policy for IAM modifications

2. IMPLEMENTATION DETAILS

Phase 1: Infrastructure & Logging

Task 1.1: Multi-Region CloudTrail

- Created CloudTrail named Security-Monitoring-Trail
- Enabled multi-region logging to capture all API activity
- Integrated with CloudWatch Logs group /aws/cloudtrail/security-trail-logs
- Configured IAM role CloudTrailRoleForCloudWatchLogs for log delivery

Log group name

/aws/cloudtrail/security-monitoring

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

IAM Role [Info](#)

AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☒ New

☐ Existing

Role name

CloudTrailRoleForCloudWatchLogs_Security-Monitoring-Trail

► Policy document

Customer managed AWS KMS key

☒ New

☐ Existing

AWS KMS alias

alias/cloudtrail-key

KMS key and S3 bucket must be in the same region.

▼ Additional settings

Log file validation

[Info](#)

CloudWatch

Favorites and recents

Ingestion

Dashboards

► Alarms

► AI Operations

► GenAI Observability

► Application Signals (APM)

► Infrastructure Monitoring

▼ Logs

Log Management [New](#)

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

► Metrics

► Network Monitoring

► Setup

Metric filter "SecurityTeam" has been created.

×

/aws/cloudtrail/security-monitoring

Actions

View in Logs Insights

Start tailing

Search log group

▼ Log group details

Log class

Standard

[Info](#)

ARN

arn:aws:logs:eu-north-1:947856325898:log-group:/aws/cloudtrail/security-monitoring:

Creation time

19 hours ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Deletion protection

☒ Off

Data protection

-

Sensitive data count

-

Custom field indexes

[Configure](#)

Transformer

[Configure](#)

Anomaly detection

[Configure](#)

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Field indexes

Transformer

Metric filters (1)

Find metric filters

Edit

Delete

Create alarm

Create metric filter

< 1 >

SecurityTeam

Filter pattern

((\${eventName} = "GetSecretValue") && (\${requestParameters.secretId} = "Production_Database_Credentials"))

Field selection criteria

-

CloudTrail > Trails

Trail successfully created

You can now enrich CloudTrail events with additional information by adding resource tags and IAM global keys in CloudTrail Lake. [Learn more](#)

Trails

Copy events to LakeDeleteCreate trail

Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Status
Security-Monitoring-Trail	Europe (Stockholm)	Yes	arn:aws:cloudtrail:eu-north-1:947856325898:trail/Security-Monitoring-Trail	Disabled	No	aws-cloudtrail-logs-947856325898-ee237ca1	-	arn:aws:logs:eu-north-1:947856325898:log-group:/aws/cloudtrail/security-monitoring*	Logging

Task 1.2: Honeytoken Secret

- Created secret in AWS Secrets Manager
- Secret name: Production_Database_Credentials
- Secret value: {"username":"admin","password":"ProjectPassword123!"}
- Enabled default encryption

AWS Secrets Manager > Secrets > Production_Database_Credentials

You successfully stored the secret Production_Database_Credentials. To show it in the list, choose Refresh. Use the sample code to update your applications to retrieve this secret.

Production_Database_Credentials

Secret details

Encryption key

aws/secretsmanager

Secret name

Production_Database_Credentials

Secret ARN

arn:aws:secretsmanager:eu-north-1:947856325898:secret:Production_Database_Credentials-Jub5EF

Secret description

-

Secret type

-

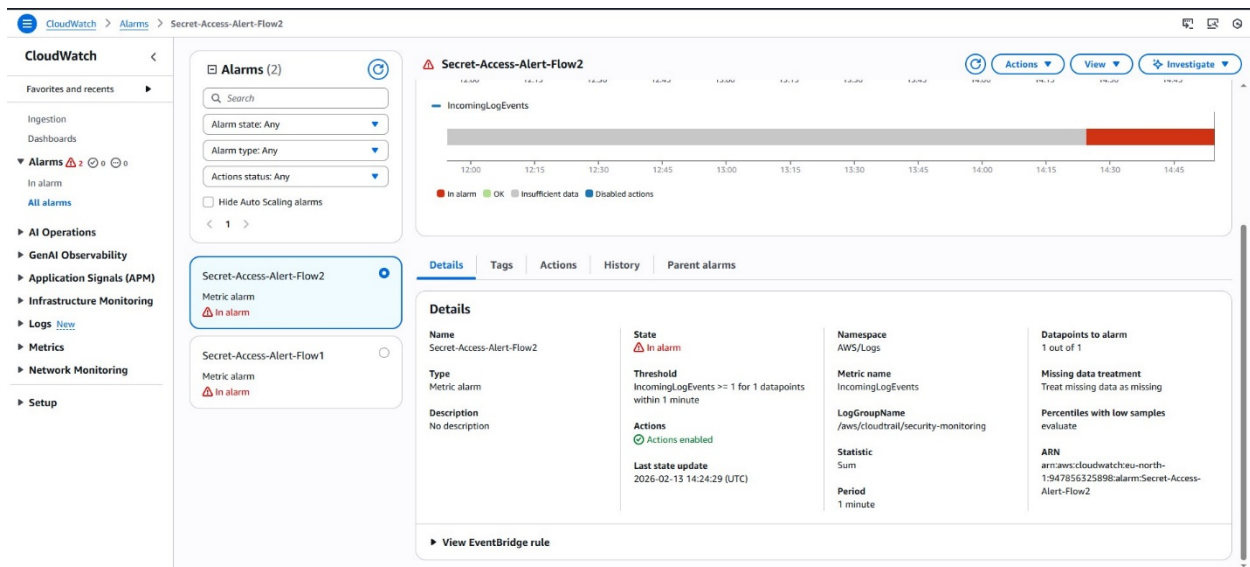
Actions

OverviewRotationVersionsReplicationTags

Secret versions (1)

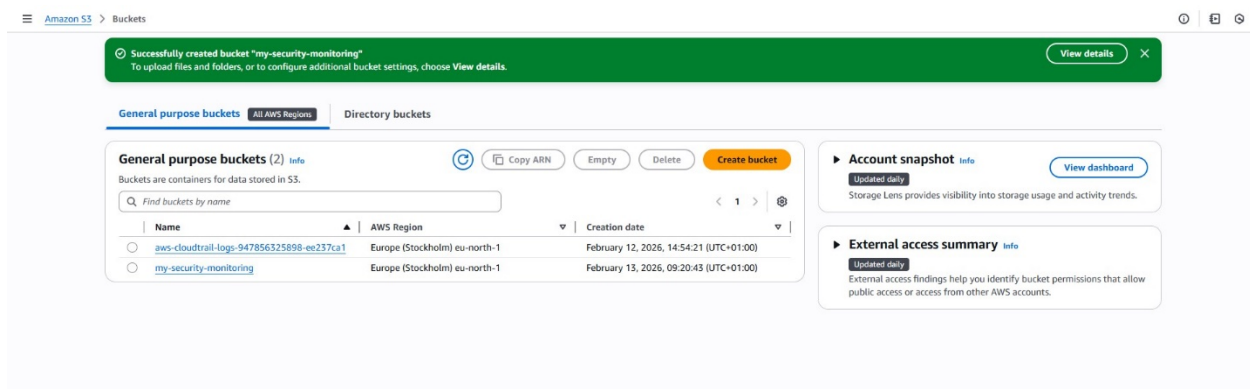
Find secret versions

Version ID	Staging labels	Last accessed	Created on (UTC)
d31192a3-24ae-4795-97e1-f64e09c7b3df	AWSCURRENT	-	February 13, 2026 at 07:05:40



Task 1.3: S3 Server Access Logging

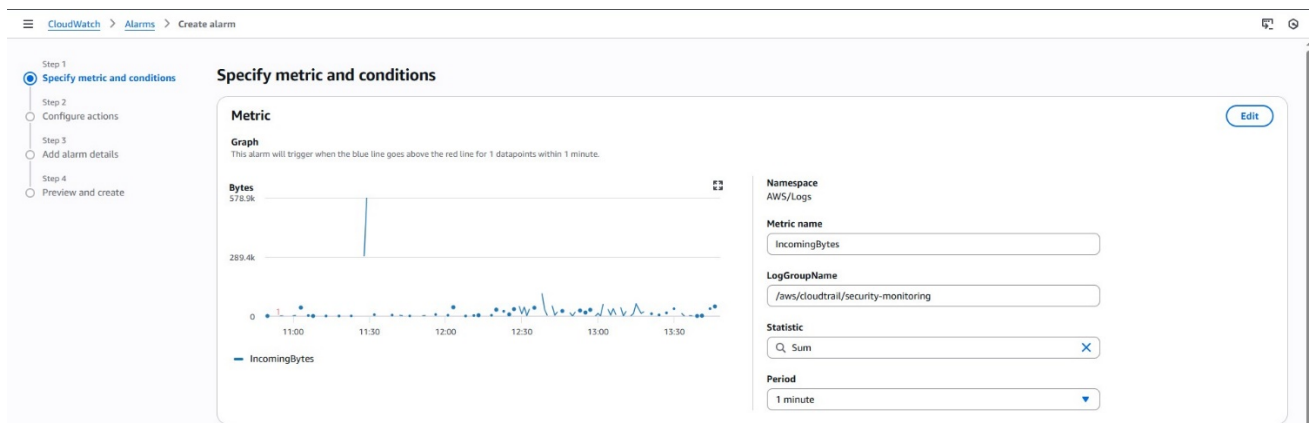
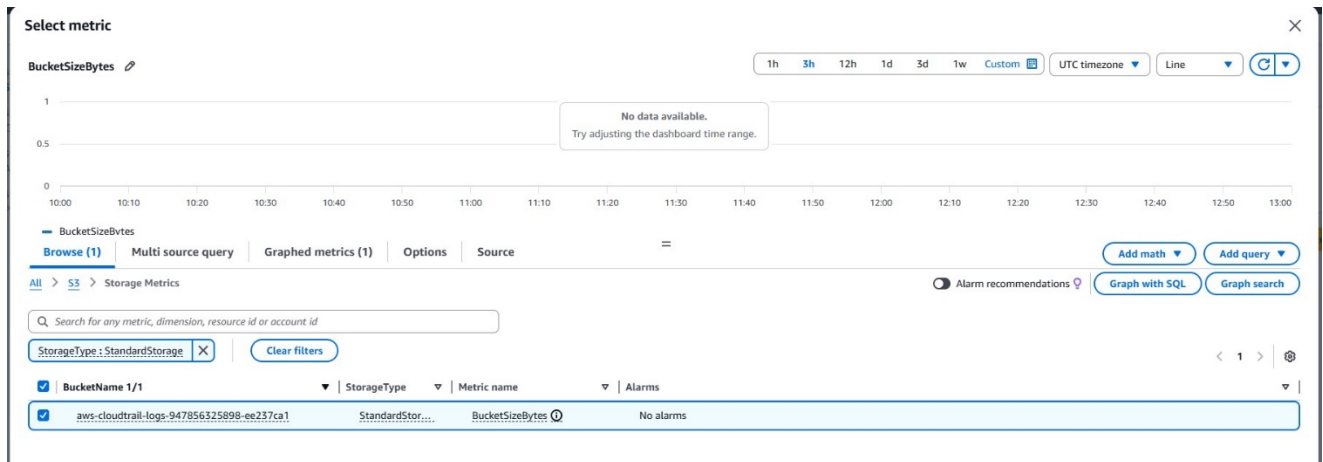
- Created source bucket: my-app-bucket-[unique-id]
- Created logs bucket: my-app-bucket-logs-[unique-id]
- Enabled server access logging with target prefix logs/



Phase 2: Detection Logic - Flow 1 (CloudWatch)

Task 2.1: Metric Filter

- Created filter in CloudWatch Logs group
- Pattern: { (\$.eventName = "GetSecretValue") && (\$.requestParameters.secretId = "Production_Database_Credentials") }
- Metric namespace: SecurityTeam/SensitiveAccess
- Metric name: ProductionSecretAccessCount



☰ AWS Secrets Manager > Secrets > Production_Database_Credentials ⓘ 📄 🔍

🔔 You successfully stored the secret **Production_Database_Credentials**. To show it in the list, choose Refresh. Use the sample code to update your applications to retrieve this secret.

View details See sample code ✕

Production_Database_Credentials

Secret details ⓘ Actions ▾

Encryption key
aws/secretsmanager

Secret name
Production_Database_Credentials

Secret ARN
arn:aws:secretsmanager:eu-north-1:947856325898:secret:Production_Database_Credentials-Jub5EF

Secret description
-
Secret type
-

Overview Rotation **Versions** Replication Tags

Secret versions (1) ⓘ Info

Version ID	Staging labels	Last accessed	Created on (UTC)
d31192a3-24ae-4795-97e1-f64e09c7b3df	AWSCURRENT	-	February 13, 2026 at 07:05:40

Task 2.2: CloudWatch Alarm

- Created alarm based on metric filter
- Statistic: Sum (critical for accurate count)
- Period: 1 minute
- Condition: ≥ 1 (alerts on single access)
- State: OK (pending initial trigger)

☰ CloudWatch > Alarms > Create alarm ⓘ 📄 🔍

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Remove

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ Create new topic

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

Secret-Access-Alert-Flow1

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

afuwajomooppeyemi@gmail.com

user1@example.com, user2@example.com

Create topic

Add notification

Lambda action

Add Lambda action

- Step 1
- Specify metric and conditions
- Step 2
- Configure actions
- Step 3
- Add alarm details**
- Step 4
- Preview and create

Add alarm details

Name and description

Alarm name

Secret-Access-Alert-Flow1

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

This is an H1
Double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024).

📘 Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [Info](#)

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 tags.

[Cancel](#) [Previous](#) [Next](#)

- Step 1
- Specify metric and conditions
- Step 2
- Configure actions
- Step 3
- Add alarm details
- Preview and create**

Preview and create

Step 1: Specify metric and conditions

[Edit](#)

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

Bytes

578.9k

289.4k

IncomingBytes

Namespace

AWS/Logs

Metric name

IncomingBytes

LogGroupName

/aws/cloudtrail/security-monitoring

Statistic

Sum

Period

1 minute

Conditions

Threshold type

Static

Whenever IncomingBytes is

Greater/Equal (>=)

than...

1

aws Search [Alt+S]

Europe (Stockholm) Opeyemi Daniel Afawajomo 014715613263880 Opeyemi Daniel Afawajomo

CloudWatch > Alarms > Create alarm

2023/06

0 11:00 11:30 12:00 12:30 13:00 13:30

IncomingBytes

LogGroupName
/aws/cloudtrail/security-monitoring

Statistic
Sum

Period
1 minute

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever IncomingBytes is...

Define the alarm condition.

☐ Greater
> threshold

☒ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

then...

Define the threshold value.

1

Must be a number.

Additional configuration

Cancel Next

Task 2.3: SNS Notification (Flow 1)

- Created SNS topic: Secrets-Alerts-Flow1
- Email subscription: [Student Email]
- Confirmed subscription via email link

- Step 1
Specify metric and conditions
- Step 2
Configure actions
- Step 3
Add alarm details
- Step 4
Preview and create

Configure actions

Notification

Alarm state trigger

Define the alarm state that will trigger this action.

☒ In alarm

The metric or expression is outside of the defined threshold.

☐ OK

The metric or expression is within the defined threshold.

☐ Insufficient data

The alarm has just started or not enough data is available.

Remove

Send a notification to the following SNS topic

Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic☒ Create new topic☐ Use topic ARN to notify other accounts

Create a new topic...

The topic name must be unique.

Secret-Access-Alert-Flow1

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...

Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

afuwajomoopayemi@gmail.com

user1@example.com, user2@example.com

Create topic

Add notification

Lambda action

Add Lambda action

AWS Notification - Subscription Confirmation Spam x

AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report not spam

You have chosen to subscribe to the topic:

arn:aws:sns:eu-north-1:947856325898:Default_CloudWatch_Alarms_Topic

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):

[Confirm subscription](#)Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to [sns-opt-out](#)

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

Metrics

Network Monitoring

Setup

Metric filter "SecurityTeam" has been created.

/aws/cloudtrail/security-monitoring

ActionsView in Logs InsightsStart tailingSearch log group

Log group details

Log class

Standard

ARN

arn:aws:logs:eu-north-1:947856325898:log-group:/aws/cloudtrail/security-monitoring:

Creation time

19 hours ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

Contributor Insights rules

KMS key ID

Deletion protection

Off

Data protection

Sensitive data count

Custom field indexes

Transformer

Anomaly detection

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Field indexes

Transformer

Metric filters (1)

EditDeleteCreate alarmCreate metric filter

Find metric filters

SecurityTeam

Filter pattern

{ (\$.eventName = "GetSecretValue") && (\$.requestParameters.secretId = "Production_Database_Credentials") }

Field selection criteria

-

ALARM: "Secret-Access-Alert-Flow2" in EU (Stockholm)

AWS Notifications

4:53 AM (5 hours ago)

You are receiving this email because your Amazon CloudWatch Alarm "Secret-Access-Alert-Flow2" in the EU (Stockholm) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (14/02/25 03:52:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Saturday 14 February, 2025 03:53:28 UTC".

View this alarm in the AWS Management Console: <https://eu-north-1.console.aws.amazon.com/cloudwatch/nextlink?region=eu-north-1#alarmsV2:alarm/Secret-Access-Alert-Flow2>

Alarm Details:

- Name: Secret-Access-Alert-Flow2
- Description:
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (14/02/25 03:52:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Saturday 14 February, 2025 03:53:28 UTC
- AWS Account: 947856325898
- Alarm Arn: arn:aws:cloudwatch:eu-north-1:947856325898:alarm:Secret-Access-Alert-Flow2

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:

- MetricNamespace: AWS/Logs
- MetricName: IncomingLogEvents
- Dimensions: [LogGroupName = /aws/cloudtrail/security-monitoring]
- Period: 60 seconds
- Statistic: Sum
- Unit: not specified
- TreatMissingData: missing

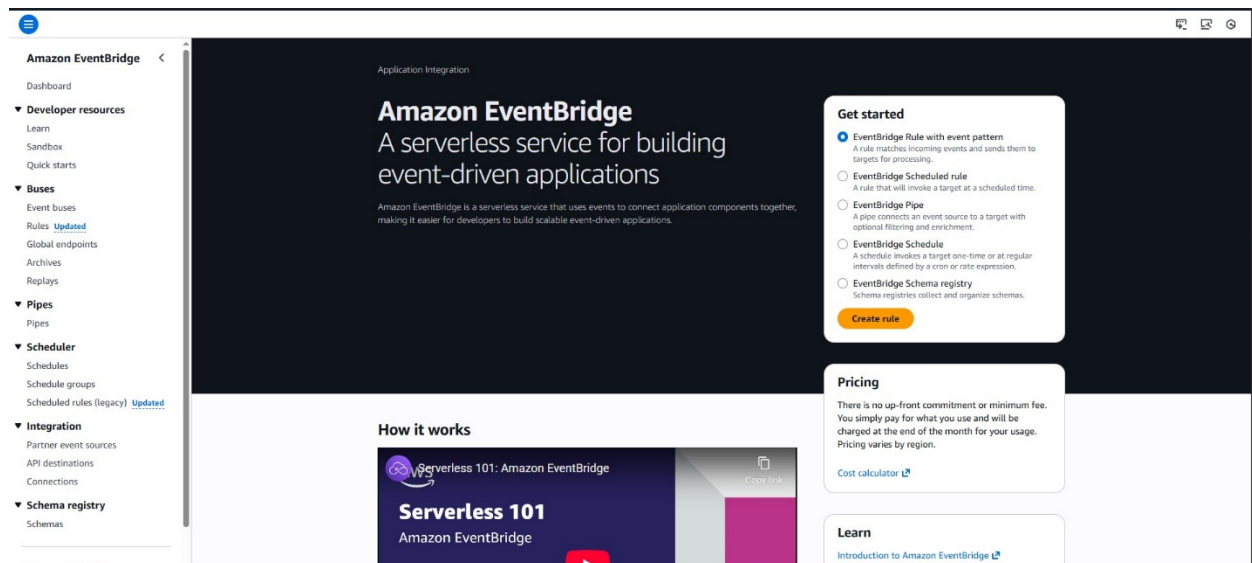
State Change Actions:

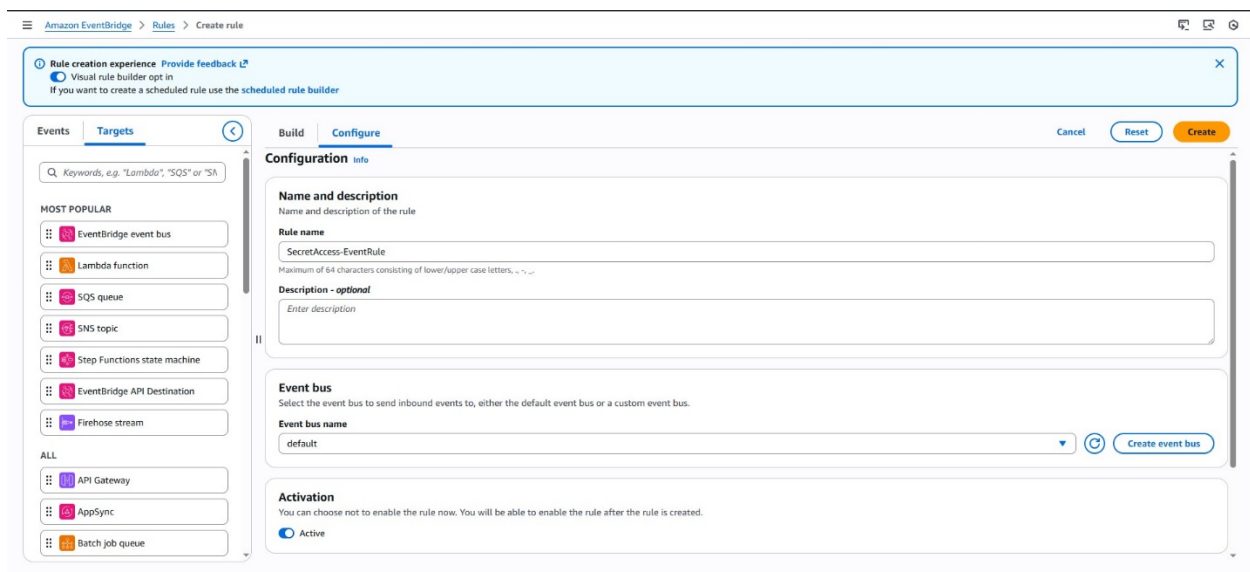
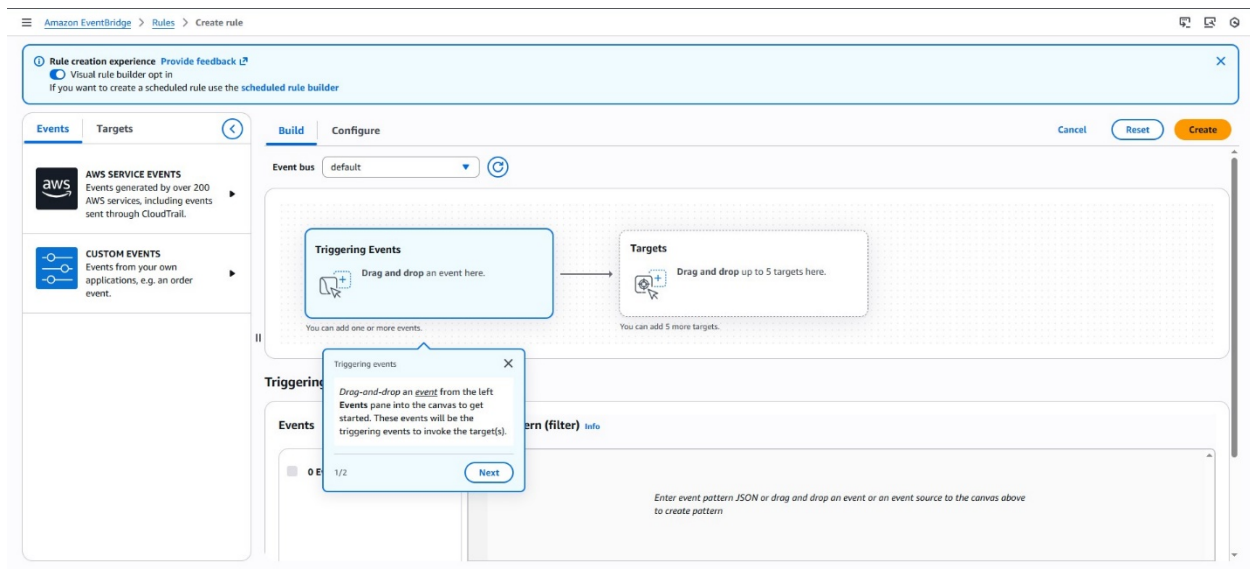
- OK:
- ALARM: [arn:aws:sns:eu-north-1:947856325898:Secret-Access-Alert-Flow2]
- INSUFFICIENT_DATA:

Phase 3: Detection Logic - Flow 2 (EventBridge)

Task 3.1-3.2: Event-Driven Rule

- Created EventBridge rule: SecretsManager-RealTime-Alert





- Event pattern:

Amazon EventBridge > Rules > SecretAccess-EventRule

Rule "SecretAccess-EventRule" was successfully created

SecretAccess-EventRule

[Edit](#) [Disable](#) [Delete](#) [CloudFormation Template](#)

Rule details		Status	Event bus name	Type
Rule name	SecretAccess-EventRule	Enabled	default	Standard
Description	-			
Rule ARN	arn:aws:events:eu-north-1:947856325898:rule/SecretAccess-EventRule	Event bus ARN	arn:aws:events:eu-north-1:947856325898:event-bus/default	

[Event pattern](#) [Targets](#) [Monitoring](#) [Tags](#)

Event pattern

```

1 {
2   "source": ["aws.secretsmanager"],
3   "detail-type": ["AWS API Call via CloudTrail"],
4   "detail": {
5     "eventName": ["GetSecretValue"],
6     "requestParameters": {
7       "secretId": ["Production_Database_Credentials"]
8     }
9   }
10 }

```

[Copy](#)

CloudWatch > Alarms > Secret-Access-Alert-Flow2

Secret-Access-Alert-Flow2

[Actions](#) [View](#) [Investigate](#)

IncomingLogEvents

Legend: In alarm (red), OK (green), Insufficient data (grey), Disabled actions (blue)

Secret-Access-Alert-Flow2		State	Namespace	Datapoints to alarm
Metric alarm	In alarm	In alarm	AWS/Logs	1 out of 1
Secret-Access-Alert-Flow1	Metric alarm	In alarm		

Details

Name	Secret-Access-Alert-Flow2	State	In alarm	Namespace	AWS/Logs	Datapoints to alarm	1 out of 1
Type	Metric alarm	Threshold	IncomingLogEvents >= 1 for 1 datapoints within 1 minute	Metric name	IncomingLogEvents	Missing data treatment	Treat missing data as missing
Description	No description	Actions	Actions enabled	LogGroupName	/aws/cloudtrail/security-monitoring	Percentiles with low samples	evaluate
		Last state update	2026-02-13 14:24:29 (UTC)	Statistic	Sum	ARN	arn:aws:cloudwatch:eu-north-1:947856325898:alarm:Secret-Access-Alert-Flow2
		Period	1 minute				

[View EventBridge rule](#)

- Created SNS topic: Secrets-Alerts-Flow2
- Email subscription confirmed

Task 3.3: The Breach Test

- Executed CLI command:

bash

aws secretsmanager get-secret-value --secret-id Production_Database_Credentials

- **Result:** Both alerts received within seconds

Phase 4: Active Defense Kill-Switch

Task 4.1: Lambda Function

- Created IAM role Lambda-KillSwitch-Role with permissions:

The screenshot shows the AWS IAM console interface. On the left, the navigation sidebar is visible with the 'Roles' link selected under 'Access Management'. The main panel is titled 'Roles (5)' and contains a table of existing roles. Below the table, there is a 'Roles Anywhere' section with three informational cards.

Role name	Trusted entities	Last activity
Amazon_EventBridge_Invoke_Sns_137360737	AWS Service: events	-
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2	16 minutes ago
AWSServiceRoleForSupport	AWS Service: support	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor	-
CloudTrailRoleForCloudWatchLogs_Security-Monitoring-Trail	AWS Service: cloudtrail	14 minutes ago

Roles Anywhere

Authenticate your non AWS workloads and securely provide access to AWS services.

- Access AWS from your non AWS workloads**
Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.
- X.509 Standard**
Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.
- Temporary credentials**
Use temporary credentials with ease and benefit from the enhanced security they provide.

IAM

Roles

Create role

Step 1
Select trusted entity

Step 2

**Step 3
Add permissions**

Step 4
Name, review, and create

Add permissions

Permissions policies (1/1117)

Choose one or more policies to attach to your new role.

Filter by Type

All types

1 match

Policy name	Type	Description
☒ AWSLambdaBasicExecutionRole	AWS managed	Provides write permissions to CloudWatc...

► Set permissions boundary - optional

Cancel

Previous

Next

IAM

Dashboard

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Access reports

IAM Identity Center

AWS Organizations

IAM Dashboard

Security recommendations

Root user has MFA

Root user has no active access keys

IAM resources

What's new

AWS Account

Quick Links

Tools

Additional information

IAM

Roles

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Access reports

IAM Identity Center

AWS Organizations

Role Lambda-KillSwitch-Role created.

View role

Roles (6)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Role name	Trusted entities	Last activity
☐ Amazon_EventBridge_Invoke_Sns_137360737	AWS Service: events	-
☐ AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2	23 minutes ago
☐ AWSServiceRoleForSupport	AWS Service: support	-
☐ AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor	-
☐ CloudTrailRoleForCloudWatchLogs_Security-Monitoring-Trail	AWS Service: cloudtrail	21 minutes ago
☐ Lambda-KillSwitch-Role	AWS Service: lambda	-

Roles Anywhere

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

X.509 Standard

Temporary credentials

- Lambda function code:

```
import boto3
import json

def lambda_handler(event, context):
    """
    Automatically revokes permissions from IAM user who accessed honeypot
    """
    print(f"Received event: {json.dumps(event)}")

    try:
        # Extract username from CloudTrail event
        user_name = event['detail']['userIdentity']['userName']
        print(f"Quarantining user: {user_name}")

        iam = boto3.client('iam')

        # Detach all managed policies
        paginator = iam.get_paginator('list_attached_user_policies')
        for page in paginator.paginate(UserName=user_name):
            for policy in page['AttachedPolicies']:
                iam.detach_user_policy(
                    UserName=user_name,
                    PolicyArn=policy['PolicyArn']
                )
            print(f"Detached policy: {policy['PolicyName']}")
```

```

# Also delete inline policies if any
inline_policies = iam.list_user_policies(UserName=user_name)
for policy_name in inline_policies['PolicyNames']:
    iam.delete_user_policy(
        UserName=user_name,
        PolicyName=policy_name
    )
    print(f"Deleted inline policy: {policy_name}")

# Optional: Add a permissions boundary for quarantine
# This prevents re-attaching policies
quarantine_policy_arn = "arn:aws:iam::aws:policy/AWSDenyAll" # Note: This policy doesn't exist by default
# iam.put_user_permissions_boundary(
#     UserName=user_name,
#     PermissionsBoundary=quarantine_policy_arn
# )

return {
    'statusCode': 200,
    'body': json.dumps(f'Successfully quarantined user: {user_name}')
}

except KeyError as e:
    print(f"Could not extract username from event: {e}")
    return {
        'statusCode': 400,
        'body': json.dumps('No IAM user found in event')
    }

except Exception as e:
    print(f"Error during quarantine: {e}")

```

Task 4.2: Connect EventBridge to Lambda

- Edited EventBridge rule SecretsManager-RealTime-Alert
- Added second target: Lambda function AutoRevokeIAMUser
- Kept existing SNS target for email alerts

5. SECURITY ANALYSIS (The "Spicy" Part)

5.1 Speed Comparison: Flow 1 vs Flow 2

Metric	Flow 1 (CloudWatch)	Flow 2 (EventBridge)
Detection Mechanism	Polling-based (metric filter)	Event-driven
Latency	Up to 1 minute (alarm period)	Near real-time (seconds)
Time to Alert	~60-90 seconds	~5-15 seconds
Time to Remediation	Not applicable	~10-20 seconds

Analysis: Flow 2 (EventBridge) is significantly faster due to its event-driven architecture. CloudWatch relies on a 1-minute metric period and alarm evaluation window, introducing inherent latency. For critical security events, **EventBridge is the superior choice.**

6. LESSONS LEARNED

6.1 Key Technical Takeaways

1. Event-Driven vs Polling Architectures

- ❖ EventBridge provides **sub-second response times** ideal for security
- ❖ CloudWatch metrics are better for **trend analysis and anomaly detection** over time
- ❖ For security incidents, **speed matters** — EventBridge wins

2. IAM Permissions Are Subtle

- ❖ Lambda functions need **exact permissions** to modify other IAM entities
- ❖ Testing with a "victim" user is essential to validate the kill-switch
- ❖ Permissions boundaries provide an elegant quarantine mechanism

3. CloudTrail Event Structure

- ❖ Understanding the JSON structure is critical for accurate filtering
- ❖ The requestParameters.secretId field contains the secret name
- ❖ UserIdentity.userName provides the identity to block

4. SNS Confirmation Requirement

- ❖ Email subscriptions must be confirmed before receiving alerts
- ❖ This is a security feature preventing unauthorized subscriptions

6.2 Challenges Encountered

Challenge	Solution
KMS alias cannot be empty	Created alias alias/cloudtrail-key
IAM role name auto-population	Selected "Existing" and created role manually
EventBridge pattern discrepancy	Triggered event first, then created rule
Lambda permissions insufficient	Added inline IAM policy to Lambda role

7. CONCLUSION

This project successfully implemented a complete security monitoring and automated response system in AWS. The dual detection paths demonstrated the tradeoffs between polling-based (CloudWatch) and event-driven (EventBridge) architectures.

The active defense kill-switch proved highly effective, automatically revoking permissions from a compromised IAM user within seconds of honeypot access. This capability would significantly reduce the blast radius of credential theft in a production environment.

The system satisfies key NIST cybersecurity framework requirements for continuous monitoring, incident detection, and automated response — core capabilities for any mature cloud security program.

8. REFERENCES

AWS CloudTrail Documentation: <https://docs.aws.amazon.com/cloudtrail/>

AWS CloudWatch Logs

Documentation: <https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/>

AWS EventBridge Documentation: <https://docs.aws.amazon.com/eventbridge/>

AWS Secrets Manager Documentation: <https://docs.aws.amazon.com/secretsmanager/>

AWS Lambda Documentation: <https://docs.aws.amazon.com/lambda/>